

Data Management Plans (DMPs)

Session Objectives

By the end of this session, you'll be able to:

- Identify data management plans (DMPs) as a way to maintain good RDM practices.
- Recognize the main questions and considerations of a DMP.
- Use the DMP Assistant to start a data management plan.
- Fill out the DMP sections on data collection, storage, ethics and legal compliance for the example research project using the DMP Assistant.

Based on what we've talked about so far, what do you think a data management plan is?

What kind of information would you put it in?

Share your answers in the chat, or raise a hand!

What is a DMP?

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- **A DMP is a document that outlines how research data will be managed during and after a project's completion.**
- It doesn't need to be perfect at the first go, and it is likely that it will need to be updated to reflect changes and pivots as your research progresses.

What is a DMP?

- Every DMP will be unique, but they aim to:
 - Help you identify services, tools, workflows, and practices that are best suited to your work
 - Save you time and headaches by anticipating bottlenecks or issues before they occur
 - Enable the preservation of your research data through purposeful planning
 - Promote the transparency and reproducibility of your research

Why Create a DMP?

Why create a DMP?

- Most people create DMPs because they were told to, either by funders, supervisors, or some other body.
- Whether you are told to or not, having a DMP is a helpful piece of research planning that will improve the quality of your research.

Funder requirements

- The [Tri-Agency RDM Policy](#) has begun rolling out the DMP requirement for certain grant opportunities
- [See here for a full list of Tri-Agency competitions that require DMPs](#) (link available on webpage)

Research group/lab requirements

- It is possible that your research lab or group may already have a data management plan in place.
- This can be at the project level, or perhaps at the group/lab level to outline best practices in things like:
 - Onboarding/offboarding new students
 - Standardized file formats for specific data types
 - Standardized documentation
 - Version control
 - Training
 - Etc

Have you ever worked on a project or with a group that used a DMP?

Share your answer in the chat, or raise a hand!

The DMP Assistant

- The DMP Assistant is a browser-based tool that helps you create a DMP using structured questions and guidance.
- It is specifically created for Canadian research projects, and features templates for specific institutions and disciplines, as well as exemplars for guidance.

The DMP Assistant

- We'll be using the Alliance Template for this program.
- Contains 21 questions across 7 topical areas in RDM.
- Each question provides guidance and links to examples/documentation.

The DMP Assistant

- Data Collection
- Documentation and Metadata
- Storage and Backup
- Preservation
- Sharing and Reuse
- Responsibilities and Resources
- Ethics and Legal Compliance

Data collection

- What types of data will you collect, create, link to, acquire and/or record?
- What file formats will your data be collected in? Will these formats allow for data re-use, sharing and long-term access to the data?
- What conventions and procedures will you use to structure, name and version-control your files to help you and others better understand how your data are organized?

Documentation and Metadata

- What documentation will be needed for the data to be read and interpreted correctly in the future?
- How will you make sure that documentation is created or captured consistently throughout your project?
- If you are using a metadata standard and/or tools to document and describe your data, please list here.

Storage and Backup

- What are the anticipated storage requirements for your project, in terms of storage space (in megabytes, gigabytes, terabytes, etc.) and the length of time you will be storing it?
- How and where will your data be stored and backed up during your research project?
- How will the research team and other collaborators access, modify, and contribute data throughout the project?

Preservation

- Where will you deposit your data for long-term preservation and access at the end of your research project?
- Indicate how you will ensure your data is preservation ready. Consider preservation-friendly file formats, ensuring file integrity, anonymization and de-identification, inclusion of supporting documentation.

Sharing and Reuse

- What data will you be sharing and in what form? (e.g. raw, processed, analyzed, final).
- Have you considered what type of end-user license to include with your data?
- What steps will be taken to help the research community know that your data exists?

Responsibilities and Resources

- Identify who will be responsible for managing this project's data during and after the project and the major data management tasks for which they will be responsible.
- How will responsibilities for managing data activities be handled if substantive changes happen in the personnel overseeing the project's data, including a change of Principal Investigator?
- What resources will you require to implement your data management plan? What do you estimate the overall cost for data management to be?

Ethics and Legal Compliance

- If your research project includes sensitive data, how will you ensure that it is securely managed and accessible only to approved members of the project?
- If applicable, what strategies will you undertake to address secondary uses of sensitive data
- How will you manage legal, ethical, and intellectual property issues?

Activity!

Let's jump into the DMP Assistant for a live demo, and then we'll start filling out some sections.

This will be our first encounter with the DMP Assistant, but as we move through the course content we'll keep revisiting it to add more information.